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Lab 1

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Task 1: Characterize the environment:

- a. Fully vs. partially observable
 - This agent is partially observable, because when he loads into the environment he cannot see the dust or the dimensions of the room, rather he has to figure it out by moving around the edges and mapping it out internally
- b. Deterministic vs. Stochastic
 - This agent is deterministic, because `next_action` is a fixed set of rules and given the same percepts and internal state it always executes the same action. For example the random agent is stochastic because he uses randomness to decide the next action whilst our agent does not.
- c. Episodic vs. Sequential
 - This agent is sequential, meaning that past actions effect the next actions. For the random agent past actions have no say in the next action because he relies on randomness to choose for him, whilst our agent is following our strategy to clean up the board and must know the previous action to make a decision on the next action.
- d. Static vs. dynamic
 - The environment is static, because it does not change without the agent moving and cleaning the dirt .While the agent moves around and can clean up dirt the environment does not change after it has loaded, the cells do not reformat and the dirt does not move around.
- e. Discrete vs. continuous
 - The environment is discrete, the agent can only take up one cell at a time and there are a finite amount of cells he can occupy. The actions are also chosen from a finite set of possible actions.
- f. Single vs. multi agent
 - This is a single agent environment, because there is only one agent acting at a time.

Task 2: Develop a strategy for the agent such that it fullfills the goal and describe this strategy in a few sentences

What must the agent achieve?

The agent must:

1. Turn on first
2. Visit every cell
3. Suck whenever it detects dirt
4. Return to starting position
5. Turn off

The agent can execute a few actions, `TURN_ON`, `TURN_OFF`, `TURN_RIGHT`, `TURN_LEFT`, `SUCK`. To know which action the agent should take we should store a couple of things.

What to store:

1. Whether the agent is turned on
2. Current direction
3. Current x, y position
4. Current state (suck, bump, etc..)

Now how can the agent execute this?

Strategy:

1. The agent moves North until he detects a bump
2. The agent then moves W until he detects a bump
3. After the agent has found a bump in both N and W directions we can be sure he is in the NW corner. We can then use the strategy we saw in class which is moving W → E until we hit a bump, then we move one cell down and move E → W. That way we can be sure that we cover the whole board
4. When the agent detects a bump whilst moving S, the agent can be sure that he has reached the bottom, and if he has covered that entire row he has finished vacuuming the board
5. We will have kept track of the current x, and y positions and the agent can then work backwards according to the current x, and y positions to get back to the starting cell.
6. When the agent has reached the starting cell it turns off

Task 4: Test your program with all three provided environments.

We tested our agent with all 3 environments, and did so multiple times. They were really helpful when it came to noticing some flaw in our logic. One bug which we found out quite late was thanks to the random_big environment. The agent wasn't checking if he had finished the last row or not which often made him miss a dirt cell, or skip checking a couple of cells, we fixed this by adding an internal flag that kept track of whether or not the entire row had been checked.

Task 6: Is your agent rational?

An agent being rational means that he makes the "correct" decision based on the information he has. By that definition our agent is indeed rational, he sweeps the entire board without skipping a cell which is the correct decision based on the information he has, because he has no knowledge if a cell has dirt on it or not before he visits them, and every time he detects dirt, he sucks which is the correct decision based on the information